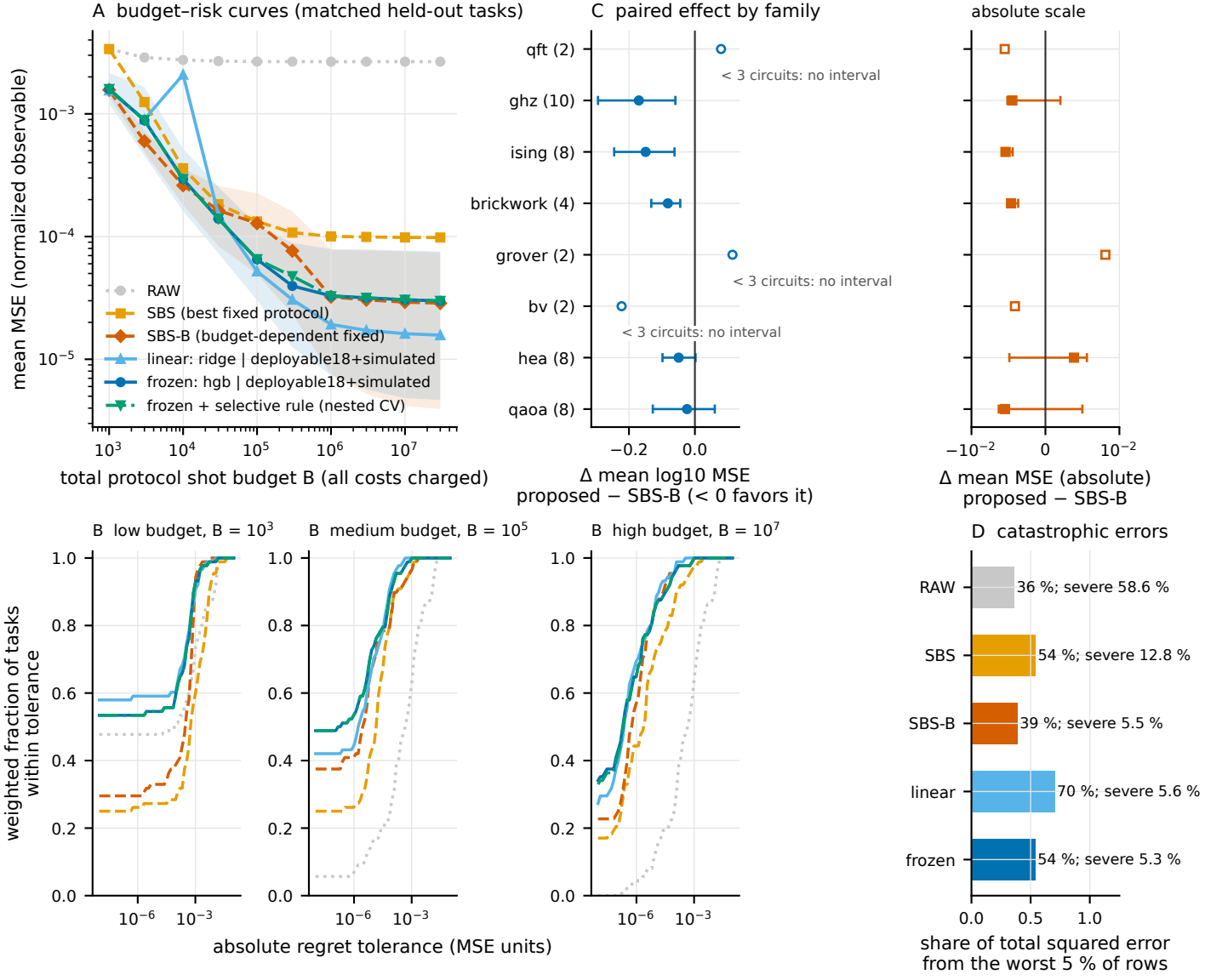
Simulated observations (emulated fake-device trials); no hardware data.
A budget–risk curves (matched held-out tasks)
mean MSE (normalized observable)
total protocol shot budget B (all costs charged)
RAW
SBS (best fixed protocol)
SBS-B (budget-dependent fixed)
linear: ridge | deployable18+simulated
frozen: hgb | deployable18+simulated
frozen + selective rule (nested CV)
C paired effect by family
qft (2)
ghz (10)
ising (8)
brickwork (4)
grover (2)
bv (2)
hea (8)
qaoa (8)
< 3 circuits: no interval
Δ mean log10 MSE
proposed − SBS-B (< 0 favors it)
absolute scale
Δ mean MSE (absolute)
proposed − SBS-B
B low budget, B = 10^3
B medium budget, B = 10^5
B high budget, B = 10^7
weighted fraction of tasks within tolerance
absolute regret tolerance (MSE units)
D catastrophic errors
RAW
SBS
SBS-B
linear
frozen
36 %; severe 58.6 %
54 %; severe 12.8 %
39 %; severe 5.5 %
70 %; severe 5.6 %
54 %; severe 5.3 %
share of total squared error
from the worst 5 % of rows